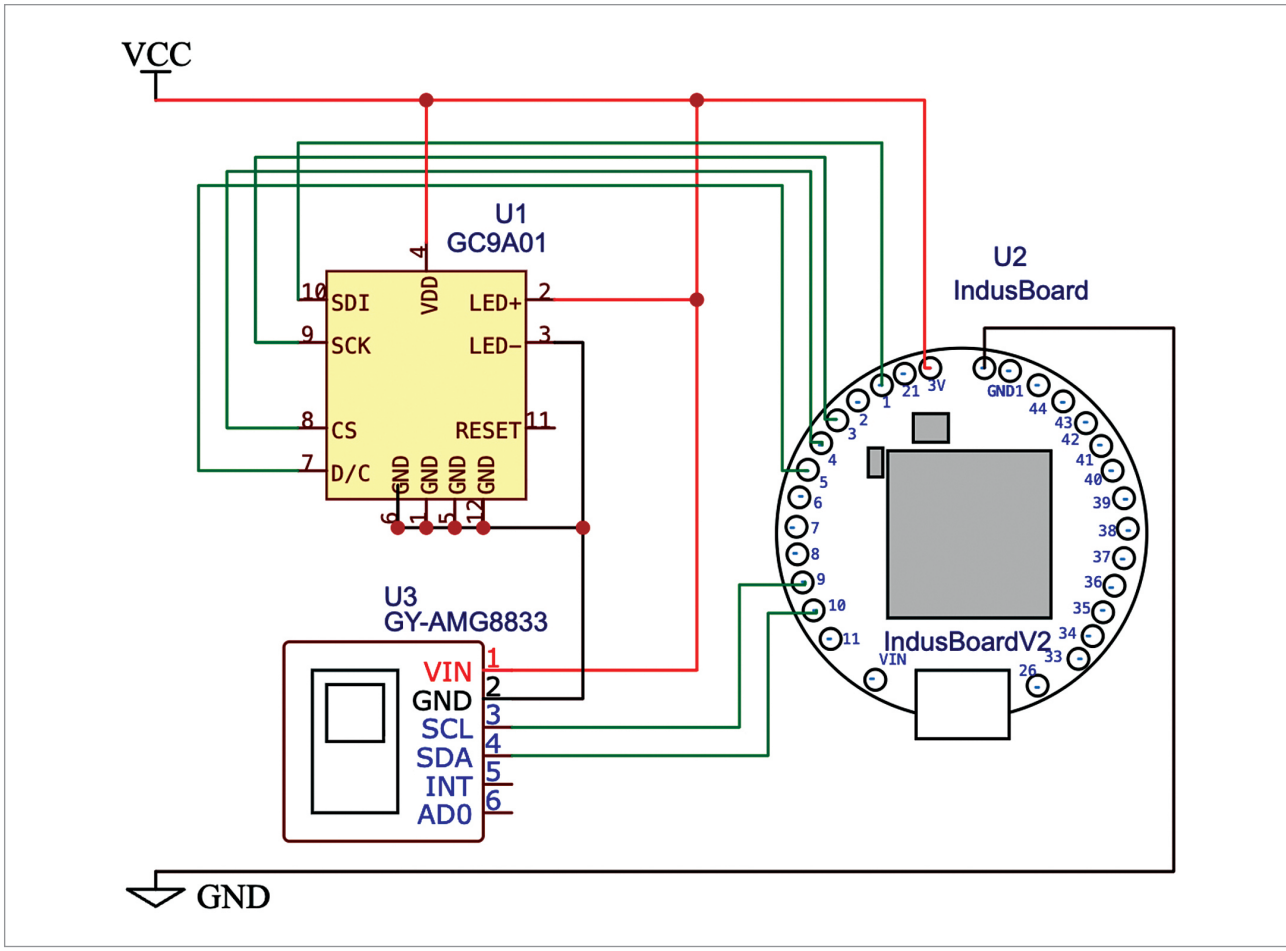


Fig. 2: Circuit of thermal camera using Indusboard coin

perform additional temperature analysis or transmit the measured data to other applications if required.

Alternatively, any other ESP32-S2-based board can be used if a compact size is not required. The corresponding circuit is shown in Fig. 3. The functionality and code remain the same; only the circuit connections need to be modified according to the ESP32-S2 board used, as shown in Fig. 3. All other components and the overall working principle remain unchanged.

As shown in Fig. 3, an ESP32-S2 board can be used to interface the GC9A01 round TFT display and GY-AMG8833 8 × 8 thermal sensor. The GC9A01 uses SPI, with SDI, SCK, CS, and D/C connected to GPIO 4, GPIO 3, GPIO 2, and GPIO 0, respectively. The AMG8833 uses I²C, with SCL and SDA connected to GPIO 9 and GPIO 8, respectively. Powered from 3.3V, the ESP32-S2 processes the 64 temperature readings into a colour-coded thermal image and displays it in real time on the GC9A01.

Construction

After uploading the source code to the IndusBoard and completing the connections, the GC9A01 round display is mounted on the top side of the IndusBoard Coin, while the AMG8833 sensor is mounted on the rear side. This keeps the overall size to 3.5cm. Alternatively, the same circuit can be built on a breadboard, as shown in Fig. 4, which shows the prototype of the thermal camera on a breadboard.

When powered, the ESP32-S2 on the IndusBoard reads the 8 × 8 temperature array from the AMG8833 over I²C, processes the data, applies colour mapping, and continuously sends the thermal image to the GC9A01 display over SPI. The result is a real-time thermal video shown on the round screen along with the centre temperature reading. Fig. 5 shows the assembled and soldered thermal camera, along with a size scale for comparison.

Software/Coding

The code is developed using the Arduino IDE with the TFT_eSPI and Adafruit AMG88xx libraries. Both libraries can be installed through Tools→Library Manager. Next, the User_Setup.h file in the TFT_eSPI library needs to be configured according to the display driver and connected GPIO pins. Since the display uses SPI, the required